



Cu/W Co-doped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ piezoelectric ceramics on structural and electrical properties

Hepeng Wang, Chao Chen*, Xiangping Jiang*, Xiaokun Huang, Xin Nie, Liwei Huang, Xing Yu

Jiangxi Key Laboratory of Advanced Ceramic Materials, School of Material Science and Engineering, Jingdezhen Ceramic University, Jingdezhen 333403, Jiangxi, China

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ABSTRACT

This article reports the significantly improved piezoelectric and ferroelectric properties of Cu/W co-doped $\text{Ca}_2\text{Bi}_2\text{Nb}_{2-x}(\text{Cu}_{1/4}\text{W}_{3/4})_x\text{O}_9$ (CBNCW- x , where $0 \leq x \leq 0.150$) high Curie temperature lead-free piezoelectric ceramics. The crystal structure, microstructure, and electric properties of CBNCW- x ceramics have been systematically analyzed. The results show that ceramic samples have a bismuth layered structure of $m=2$, which is a single-phase of CBN. The ceramic grains show the typical plate-like structure of the bismuth layer. Cu/W co-doped CBN ceramics increased the distortion of NbO_6 octahedron, and significantly enhanced the piezoelectric constant and the remnant polarization. When $x = 0.075$, it shows better comprehensive properties with an excellent d_{33} of 13.8 pC/N, a high remnant polarization of 12.10 $\mu\text{C}/\text{cm}^2$, the T_c of 918 °C, a dielectric loss value of 0.57%, and the Q_m value of 7099. Moreover, CBNCW-0.075 ceramic simple maintains the d_{33} of 12.7 pC/N after annealing 900 °C for 30 min, which shows good high-temperature stability. It suggests that the Cu/W co-doped CBN ceramics have good application prospects in the high-temperature piezoelectric field.

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1. Introduction

At present, the widely used piezoelectric material is PZT-based piezoelectric ceramics, but its Curie temperature is low ($T_c < 400$ °C), and it is a Pb-containing material, which cannot be used normally in a high-temperature environment [1–8]. Therefore, it is necessary to explore piezoelectric materials that are more suitable for high-temperature environments.

Bismuth layered structure compounds (BLSFs) are important high-temperature lead-free piezoelectric ceramic materials. They have the advantages of high Curie temperature ($T_c > 500$ °C), good thermal stability, and high quality factor, which shows a certain potential for high-temperature applications. The chemical formula of BLSFs is $(\text{Bi}_2\text{O}_2)^{2+}(\text{A}_{m-1}\text{B}_m\text{O}_{3m+1})^{2-}$, where m is the number of octahedral pseudo-perovskite layers, generally between 1 and 5; A is +1, +2, +3 valence ions or twelve-coordinated ion groups; B is a six-coordinated transition metal ion [7]. Due to the alternating arrangement of the $(\text{Bi}_2\text{O}_2)^{2+}$ layer and the $(\text{A}_{m-1}\text{B}_m\text{O}_{3m+1})^{2-}$ layer, the rotation of the spontaneous polarization (P_s) is restricted to the a - b plane, resulting in a relatively low piezoelectric performance [9–11].

Since there is almost no displacement of the lattice sites along the c -axis, the contribution of the piezoelectric and ferroelectric for the BLSFs comes from the tilt of the oxygen octahedron along the a - b plane [9]. Reports have shown that the remnant polarization can be significantly increased by increasing the oxygen octahedral distortion [9,12]. In addition, the bismuth layered piezoelectric ceramics will cause Bi_2O_3 to volatilize during the high-temperature sintering process, causing structural defects and further deterioration of its electrical properties.

$\text{CaBi}_2\text{Nb}_2\text{O}_9$ (CBN) is a typical $m=2$ bismuth layered piezoelectric ceramic with an ultra-high Curie temperature ($T_c \sim 940$ °C). It has shown a great application value in the field of high-temperature piezoelectricity. However, its piezoelectric constant is only 5 pC/N [13]. Therefore, scholars have carried out extensive research on the improvement of its electrical properties. For example: H. Chen et al. prepared $\text{Ca}_{0.95}\text{Sm}_{0.05}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics by a texturing method to significantly increase d_{33} to 23 pC/N [14]. H. Chen et al. compared the effects of acceptor doping and donor doping on the conductivity of CBN ceramics [15]. H. Chen et al. used a two-step sintering method to prepare CBN ceramics with a d_{33} of 13.5 pC/N [13]. And the modified way with a substitution at A -site, B -site, and A/B site, which include Ce/W ions modified CBN ceramics by doping at A/B site that improved d_{33} to 13.4 pC/N [16]. In the research, B -site doping can maintain a high Curie temperature, while effectively improving

* Corresponding authors.

E-mail addresses: cc2762@163.com (C. Chen), jiangxp64@163.com (X. Jiang).

the electrical performance of CBN piezoelectric ceramics [17–24]. Examples are as follows: Chen et al. [17] used W/Cr ions to do the B-site composite doping of CBN ceramics, and the piezoelectric constant was significantly increased from 5 to 15 pC/N, Curie temperature of 931 °C; Xing et al. [18] used W^{6+} to do the B-site doping of CBN ceramics, which significantly increased the resistivity and breakdown strength of CBN ceramics, and the piezoelectric constant was increased to 13 pC/N. In addition, Li et al. [25] used Cu^{2+} and Nb^{5+} co-doped $Bi_4Ti_3O_{12}$ ceramics, and the piezoelectric constant was significantly increased from 8 to 38 pC/N. To sum up those examples, doping with W^{6+} can effectively improve the electrical properties of piezoelectric ceramics, and CuO has the advantages of lowering the sintering temperature, improving the quality factor, and reducing the dielectric loss [26,27]. However, the co-doping modification of Cu^{2+} and W^{6+} in the CBN ceramics remains to be studied. Therefore, this paper adopts the traditional solid-state method to prepare Cu/W ions co-doped CBN piezoelectric ceramics, and hopes for obtaining a high-temperature piezoelectric ceramic with improved piezoelectric constant, reduced dielectric loss, and increased quality factor.

2. Experimental process

Cuprum and tungsten co-doped $CaBi_2Nb_2O_9$ ceramics with chemical formula $Ca_2Bi_2Nb_{2-x}(Cu_{1/4}W_{3/4})_xO_9$ (CBNCW- x , $x = 0.000, 0.025, 0.075, 0.125$, and 0.150) were prepared by conventional ceramic sintering process using high-purity raw materials [Bi_2O_3 (99.999%), $CaCO_3$ (99%), Nb_2O_5 (99.5%), CuO (99%), WO_3 (99.99%)]. The raw materials were weighed and mixed in a nylon jar with high-purity alcohol for 24 h. The dried mixture was calcined at 750 °C for 3 h and then re-milled for 24 h. After drying, the powders were granulated with polyvinyl alcohol (PVA, 5 wt%) binder, and then aged for 24 h. Disks with a diameter of 13 mm and thickness of 1 mm were formed by pressing at 300 ± 20 MPa, then burned out the PVA at 650 °C for 2.5 h. The pressed pellets were sintered between 1030 and 1150 °C for 2 h with 5 °C/min. The sintered ceramics were polished to about 0.5 mm thick, and then painted with Ag/Pd paste and fired at 800 °C for 10 min with 10 °C/min. The ceramics were poled in silicone oil at 180 °C for 30 min with a direct-current electric field of 200–260 kV/cm.

The XRD patterns of the sintered ceramics were determined at room temperature using an X-ray diffractometer (D8 Advanced Germany) equipped with Cu-K α radiation. The Raman spectra were measured at room temperature by a Raman spectrometer (LabRAM HR800) equipped with a laser of 532 nm wavelength. The micro-structure of the sintered ceramics with thermally etched was characterized by a field-emission scanning electron microscope (SU8010, Japan). The temperature dependence of dielectric constant and loss of ceramics was measured by an LCR analyzer (TH2828s, Tong Hui, China). The ferroelectric hysteresis of ceramics was measured at 10 Hz by using a ferroelectric meter (Trek model 609B). The piezoelectric constants of ceramics were measured by using a quasi-static d_{33} meter (ZJ-3A, China). The resonance frequency (f_r) and antiresonance frequency (f_a) of polarized ceramics were determined by using an LCR analyzer (HP4294A, Agilent, US).

3. Results and discussion

First, the structures of the ceramic samples are characterized. Fig. 1(a) shows the X-ray diffraction (XRD) spectrum of the CBNCW- x ceramic samples measured from 20° to 70°. The number and

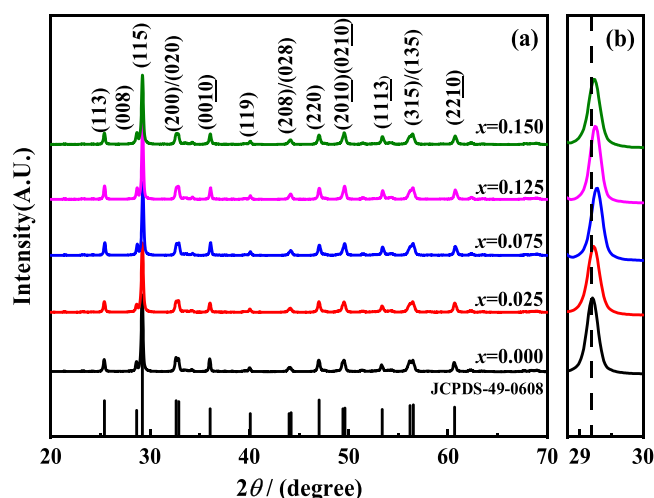


Fig. 1. (a) The XRD patterns of CBNCW- x ceramic samples and (b) enlarged view of (115) characteristic peak.

position of the diffraction peaks of each component sample are almost similar, and index according to the CBN crystal phase (JCPDS 49-0608). The highest intensity diffraction peak (115) is consistent with the strongest diffraction peak of bismuth layered structure (112 $m+1$) [9]. No secondary phase was detected within the scan range, indicating that the doped Cu^{2+} and W^{6+} are diffused into the CBN lattice, and forming a single-phase compound [28]. As seen from Fig. 1(b), the (115) characteristic peak shifts to the higher angle direction and then to the lower direction with the doping amount x increases. On the one hand, the Cu/W composite ion radius is smaller than that of Nb^{5+} [$r(Nb^{5+}) = 0.064$ nm; $r(Cu^{2+}) = 0.073$ nm, $r(W^{6+}) = 0.06$ nm, $r(Cu/W) = 0.063$ nm], resulting in a shrinkage in the lattice, and making the diffraction peak shift in the higher angle direction. On the other hand, the Cu/W doping significantly reduces the sintering temperature and Bi_2O_3 volatilization. As a result, there are less lattice defects, and the unit cell size becomes larger, which causes the diffraction peak to shift to a lower angle direction. At a relatively low doping level, the reduced ion radius is the main factor for the shift of the diffraction peak. With the increase of the doping level, the decrease of lattice defects gradually becomes the main factor for the shift of the diffraction peak.

To explore the microscopic crystal structures of the CBNCW- x ceramic samples, the XRD patterns were Rietveld refined by the GSAS-EXPGUI software in the $A2_1am$ space group. As shown in Fig. 2, the fitted curves of all components almost overlap with the original curves. The reliability factors R_{wp} , R_p , and χ^2 are important parameters to measure whether the refined fitting results are accurate. The R_{wp} (8.29–9.94%), R_p (5.94%–7.09%), and χ^2 (4.954–7.06) are reasonable, indicating that the refined results are sufficient credibility.

Fig. 3(a) shows the refined values of the lattice parameters a , b , and c of the CBNCW- x ceramics. The lattice parameters a , b , and c all decrease first and then increase with the increase of the doping level x . Fig. 3(b) shows the dependence of the cell volume and a/b on the doping amount of the CBNCW- x ceramics. Among them, the cell volume first decreases and then increases with the increase of the doping amount x , which is consistent with the shift of the (115)

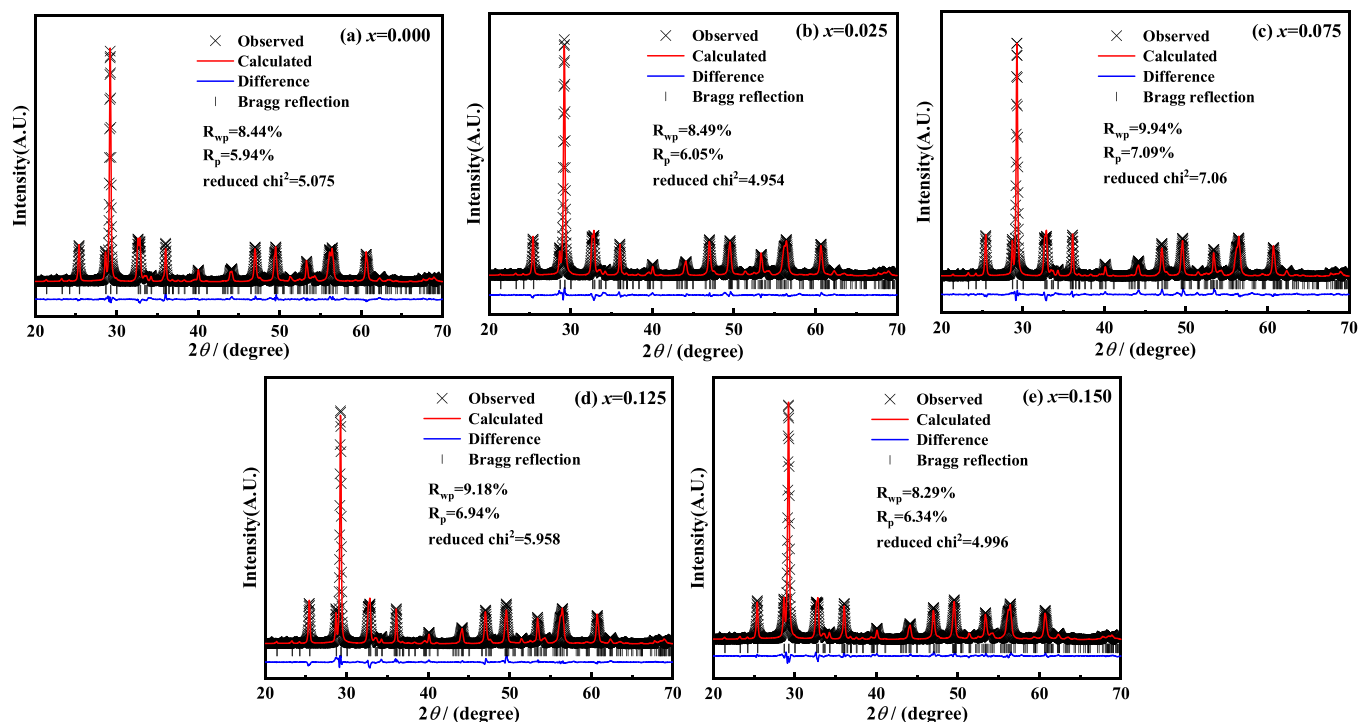


Fig. 2. The Rietveld refined XRD patterns of CBNCW-x ceramic samples.

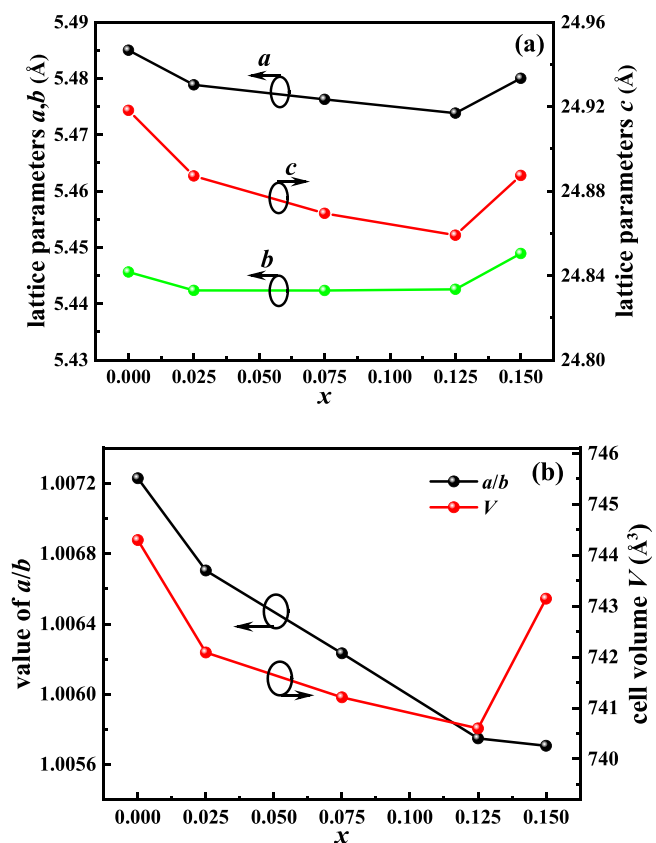


Fig. 3. The Rietveld refined data of CBNCW-x ceramic samples: (a) lattice parameters and (b) cell volume and value of a/b .

characteristic peak. For the BLSFs compounds, the a/b can be expressed as the degree of orthogonal distortion. The larger the difference between a and b , the weaker the lattice symmetry and the greater the degree of orthogonal distortion [29]. The results show that the value of a/b decreases with the increase of the doping amount x , which indicates that Cu/W ions doping reduces the degree of orthogonal distortion of CBN ceramics.

Fig. 4(a) shows the Raman spectra of CBNCW-x ceramic samples at room temperature. The Raman curves of samples are similar to other researchers' reports [22]. To display the changes of Raman peaks intuitively and accurately, the peaks fitting of Raman curves were performed with Lorentz function and obtained 17 decomposition peaks (ν_1 to ν_{17}). Taking the pure CBN ceramic as an example, the fitting curve (red) and the original curve (black) almost completely overlap, indicating that the position of the decomposition peaks and the corresponding Raman peaks are consistent. Raman peaks of obvious intensity have been marked, wherein, the peaks of ν_6 , ν_7 , and ν_{12} changed significantly with Cu/W ion doping, while the peaks of ν_1 , ν_9 , ν_{14} , and ν_{16} did not. The peaks of ν_6 and ν_7 represent the vibration of Ca^{2+} at A-site, while the peak of ν_{12} is related to the vibration of the BO_6 octahedron [23,30–33]. Fig. 4(b) and 4(c) show that with the increase of doping amount x , the ratio of normalized intensity of ν_7 and ν_6 peaks decreases continuously, which is consistent with the decrease of lattice distortion of CBN ceramics [23,31]. The mode of ν_{12} shifts to the low wavenumber direction as the doping amount x increases, and its Raman intensity also decreases. This is due to the larger mass of Cu/W ions entering the vacancies of Nb^{5+} ($M_{Cu/W} = 153.74$ g/mol, $M_{Nb} = 92.91$ g/mol) [31].

To discover the influence of Cu/W ions diffused into Nb^{5+} vacancies in the BO_6 octahedron. During the Rietveld refinement, we have been revised the atomic coordinate parameters of the XRD data and obtained the visualized structure diagrams, which were

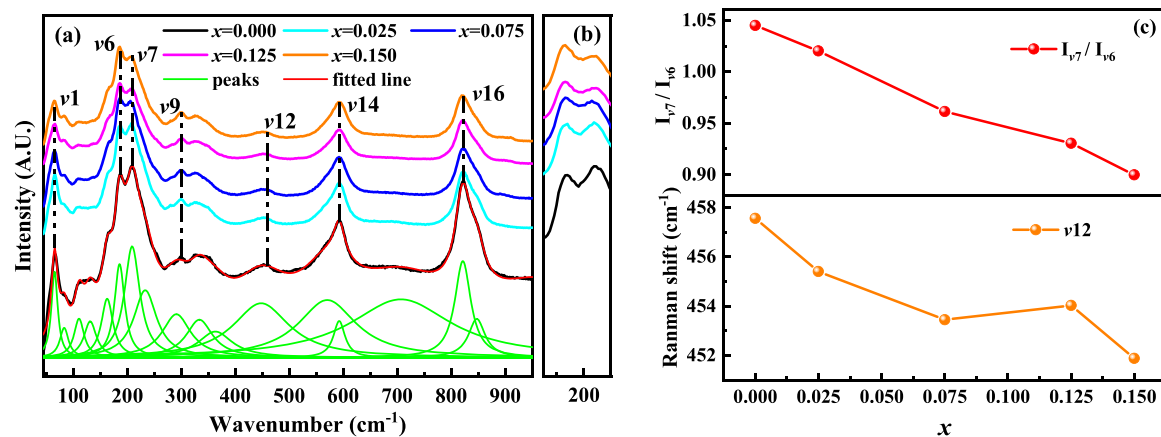


Fig. 4. (a) Raman spectrum of CBNCW-x, (b) enlarged picture near 200 cm⁻¹, and (c) part of Raman peaks.

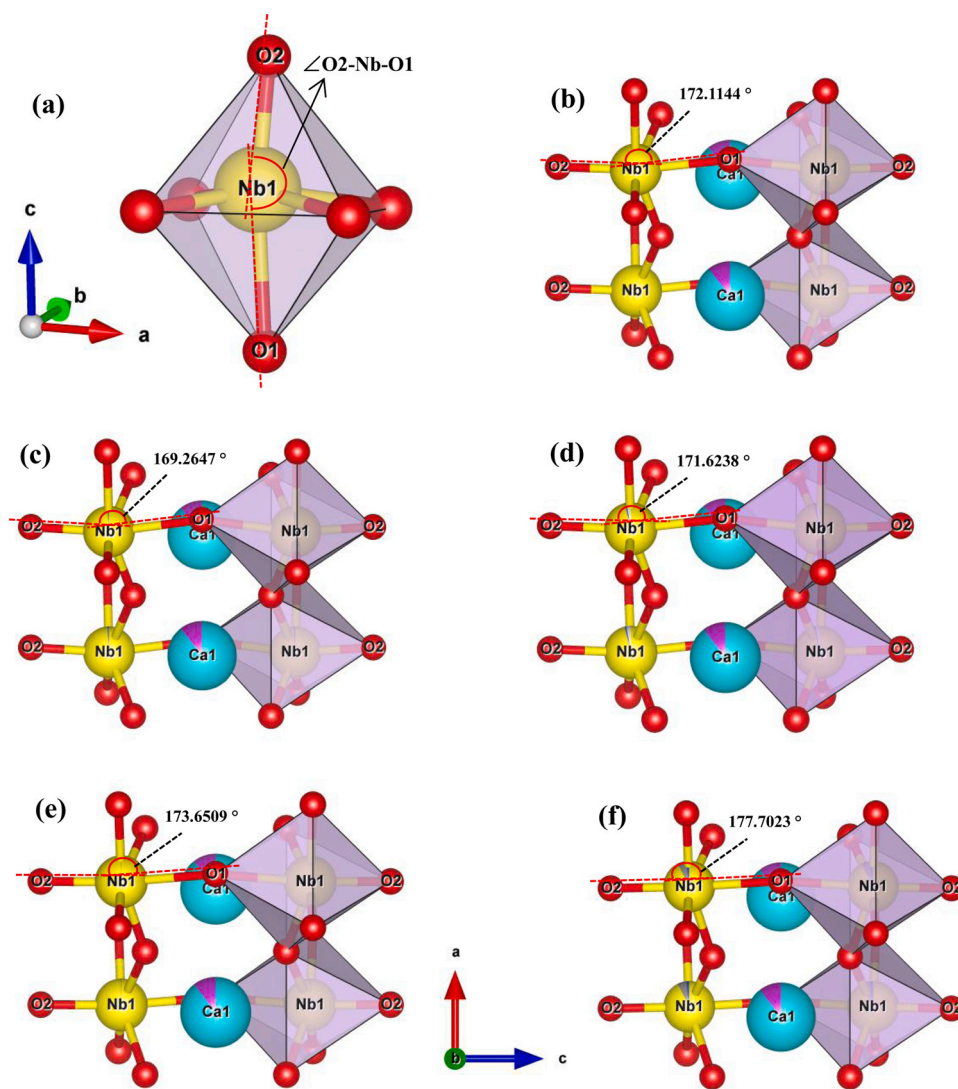


Fig. 5. The visualized crystal structures of CBNCW-x ceramics: (a) NbO₆ octahedron, (b) x = 0.000, (c) x = 0.025, (d) x = 0.075, (e) x = 0.125, and (f) x = 0.150.

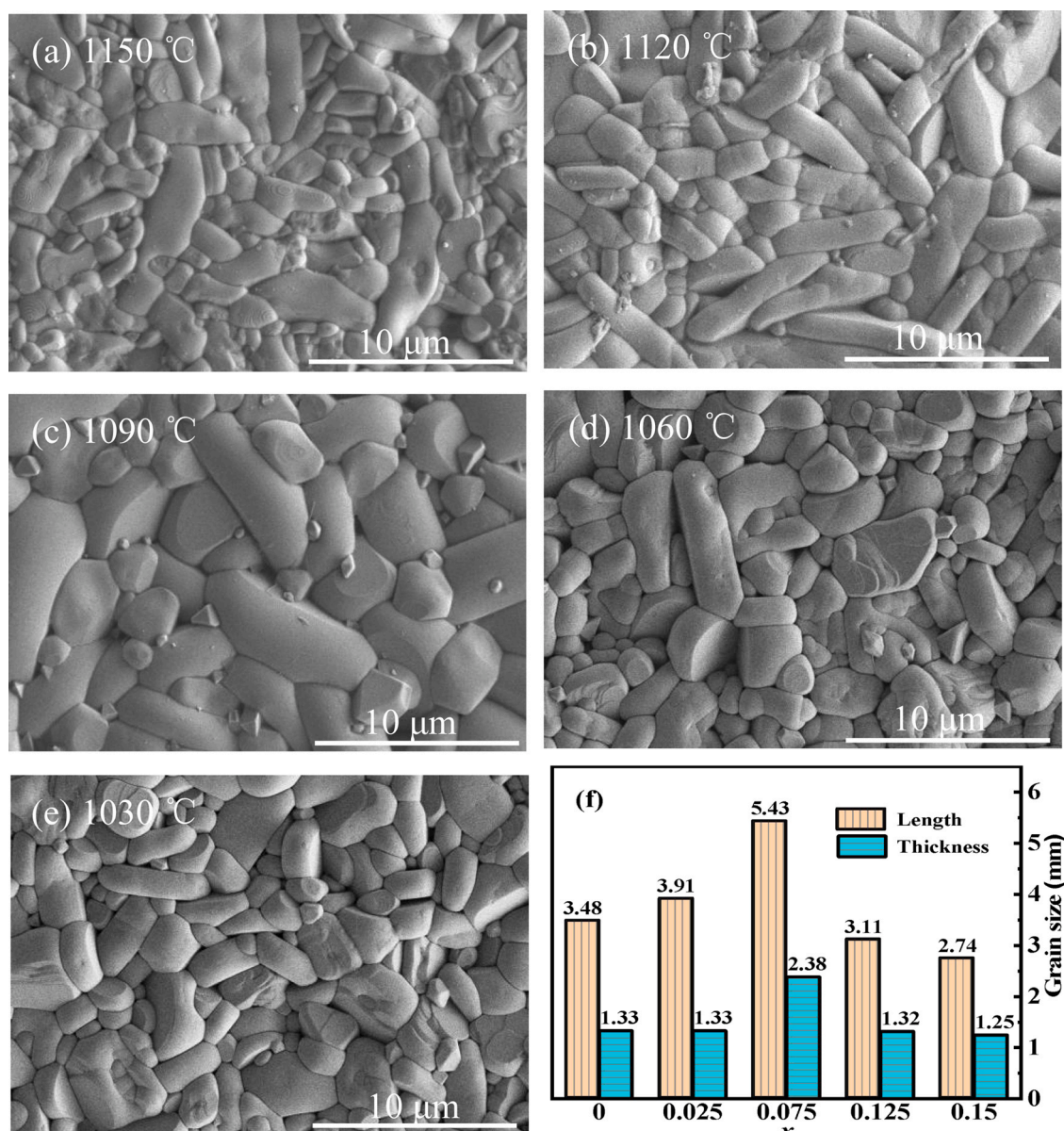


Fig. 6. The SEM pictures of CBNCW- x ceramic samples: (a) $x = 0.000$, (b) $x = 0.025$, (c) $x = 0.075$, (d) $x = 0.125$, (e) $x = 0.150$, and (f) the average grain size.

analyzed by VESTA software. Fig. 5(a) shows the structure diagram of oxygen octahedron distortion. The value of $\angle\text{O2-Nb-O1}$ shows the distortion of the NbO_6 octahedron along a - b plane, which represents the effects of crystal structure on the spontaneous polarization [9]. As shown in Fig. 5(b)-(f), Cu/W ions doping significantly affects the distortion of the NbO_6 octahedron of CBN ceramics. The value of $\angle\text{O2-Nb-O1}$ first decreases and then increases, indicating that proper doping can effectively increase the distortion of the NbO_6 octahedron. The increased distortion is conducive to the separation of positive and negative charge centers, resulting in enhanced polarization of CBN ceramics.

Fig. 6(a)-(e) show the micromorphology of thermally-etched surfaces of CBNCW- x ceramic samples. The crystal grains of ceramic samples show typical bismuth layered plate-like structure because the pseudo-perovskite structure grows faster on the a - b plane than that on the c -axis. The surface of ceramic samples shows clear grains and no obvious pores, indicating that the experimental sintering temperature is appropriate (1030–1150 °C). However, there are some bulges in the picture, especially in Fig. 6(c), the cause of which is still unknown, and similar phenomena have been reported in other reports [34,35]. Fig. 6(f) shows the average grain sizes of the ceramics, which is counted by using Nano Measurer software. When $x \leq 0.075$,

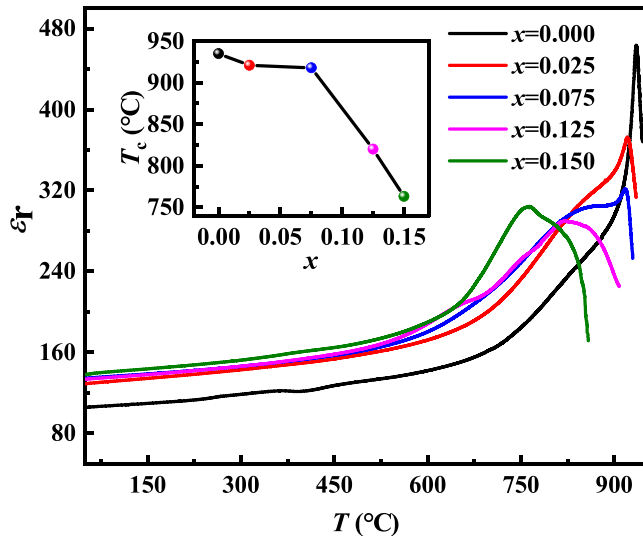


Fig. 7. The relationship between dielectric constant and temperature of CBNCW- x ceramics at 1 MHz.

the average grain sizes of the ceramic samples gradually increase, mainly due to the Cu/W ions can effectively reduce the sintering temperature. More liquid phases are formed during ceramic sintering, which promotes the growth of grains. A similar phenomenon has been observed in other papers that have been reported [26,27]. However, when $x > 0.075$, the average grain sizes gradually decrease, which may be due to the large reduction of the sintering temperature, and the influence of temperature on grain growth is more significant than that of additives. The liquid phase formed during sintering is reduced, the liquid phases cannot be connected, and a new solid-liquid boundary is formed between grain boundaries, which consumes the energy of grain boundary movement and inhibits the growth of grains.

Fig. 7 shows the curves of dependence of dielectric constant and temperature of CBNCW- x ceramic samples at 1 MHz. The dielectric peak gradually moves towards low temperature with the increase of doping content x , corresponding to the decrease of Curie temperature (T_c). Among them, the T_c of CBNCW-0.075 ceramic remained at a high level of 918 °C. The T_c is mainly related to the lattice distortion of the material that the higher the degree of distortion, the higher the T_c . The degree of lattice distortion can be expressed by the tolerance factor t . The tolerance formula is shown in Eq. (1):

$$t = \frac{r_A + r_O}{\sqrt{2}(r_B + r_O)} \quad (1)$$

where r_A , r_B , and r_O represent the radius of metal cations at A and B sites and the radius of O^{2-} , respectively [36]. As the equivalent radius of Cu/W ion is smaller than Nb^{5+} , the tolerance factor t increase, corresponding to lower lattice distortion (lower T_c), which is consistent with the XRD refinement results that the degree of orthogonal distortion keeps decreasing. With the increase of x , the dielectric constant peak of the ceramic sample gradually broadens,

showing a relaxation effect, which is due to the increased disorder of lattice cation arrangement [37,38].

To clearly represent the dielectric relaxation phenomenon of the CBNCW- x ceramic samples, the modified Curie-Weiss formula is used to characterize their dielectric relaxation. The formula is shown in Eq. (2):

$$\frac{1}{\varepsilon_r} - \frac{1}{\varepsilon_m} = C^{-1}(T - T_m)^\gamma \quad (2)$$

where ε_r is the relative dielectric constant, ε_m is the peak value of the dielectric constant, C is the Curie constant, T is the temperature, T_m is the temperature corresponding to ε_m , and γ is the dispersion factor. As shown in Fig. 8, with the increase of the doping amount x , the γ gradually increases from 1.505 to 1.677, indicating that the relaxation of the CBNCW- x ceramics gradually increases with the increase of doping level.

Fig. 9 shows the temperature dependence of the dielectric loss of the CBNCW- x ceramics at 1 MHz. The dielectric loss of all modified ceramics in the temperature range from room temperature to 500 °C is lower than that of pure CBN because the additive of CuO can effectively reduce the dielectric loss of materials [26,27]. Cu/W ions doping reduces the oxygen vacancy concentration (as shown in Eqs. 3 and 4), reducing the concentration of carriers and resulting in a lower dielectric loss. The dielectric loss of CBNW-0.075 is only 5.38% at 500 °C, indicating that CBNW- x ceramics have a good application prospect in a high-temperature lead-free piezoelectric field.

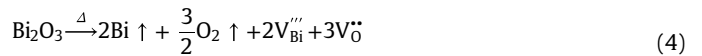
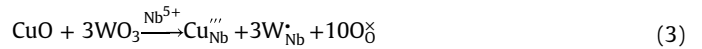


Fig. 10 shows the ferroelectric loops of the ceramic samples measured at 150 °C and 10 Hz. As shown above, all the modified samples show a saturation loop, and the remanent polarization (P_r) of the samples increases first and then decreases with the increase of Cu/W ions doping level, which corresponds to the distortion trend of NbO_6 octahedron. The improvement of ferroelectric performance can be attributed to the following aspects. Firstly, the NbO_6 octahedral distortion is enhanced at a proper doping amount, which is conducive to the separation of positive and negative charge centers and improves the polarization strength of CBN ceramics [9,12]. Secondly, the entry of Cu^{2+} and W^{6+} into Nb^{5+} vacancies is microchemical non-equivalent, and form negative and positive charge centers at the substitution points, respectively, forming electric dipole moments inside the ceramics, thus improving their ferroelectric properties [22]. Finally, due to the decrease of oxygen vacancy concentration, the pinning effect is weakened, and the reversal of the ferroelectric domain becomes easier [10]. However, the reduction of the P_r is mainly due to the reduction of oxygen octahedral distortion. When $x = 0.075$, the P_r is significantly increased from 3.12 to 12.10 $\mu C/cm^2$, indicating that Cu/W doped CBN ceramics provide a better choice for high-temperature ferroelectric materials.

Fig. 11(a) shows the dependence of d_{33} and x of the CBNCW- x ceramic samples. As the doping amount x increases, the d_{33} first increases and then decreases. As shown in Table 1., the main

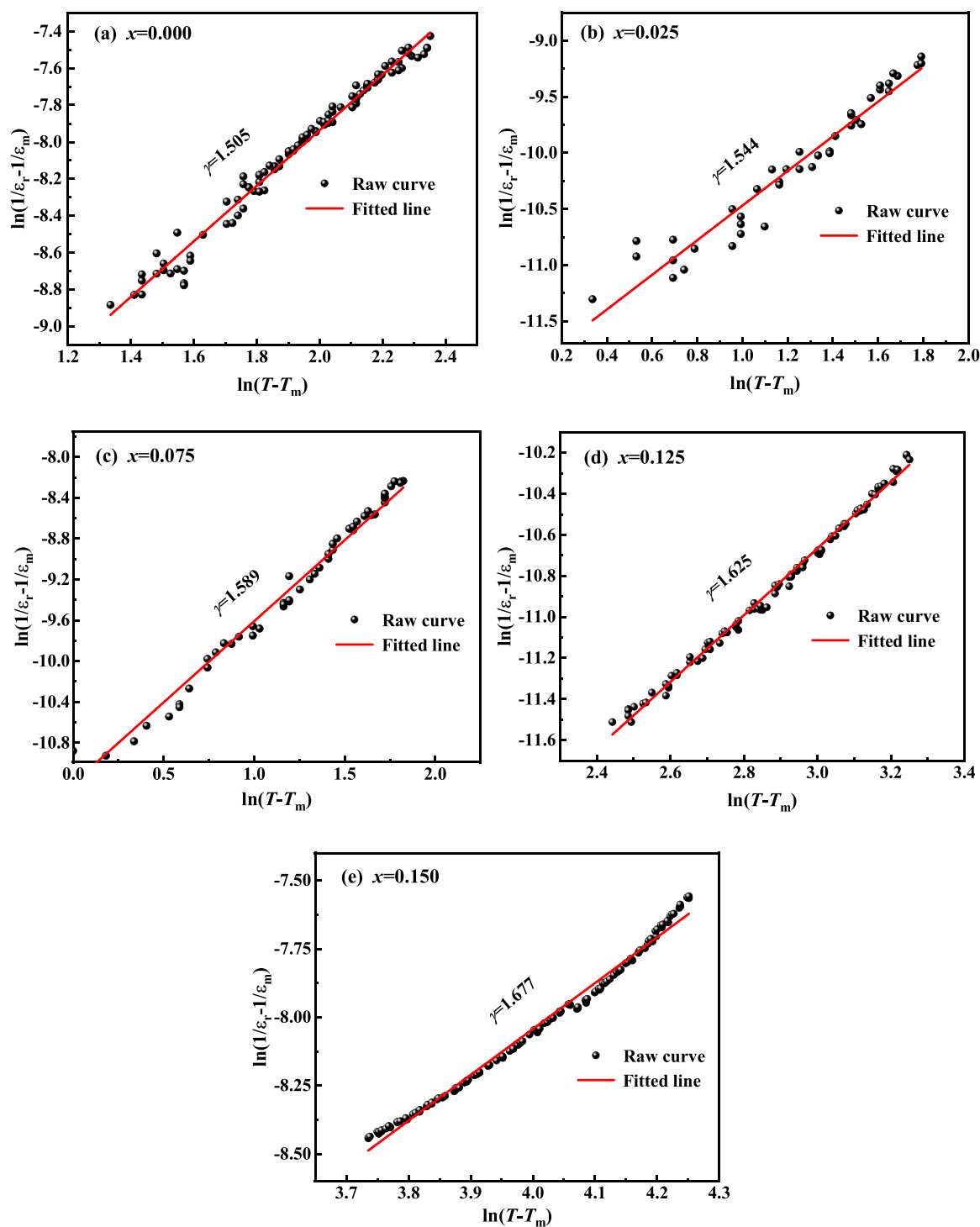


Fig. 8. The relationship between $\ln(1/\epsilon_r - 1/\epsilon_m)$ and $\ln(T - T_m)$ of CBNCW- x ceramic samples.

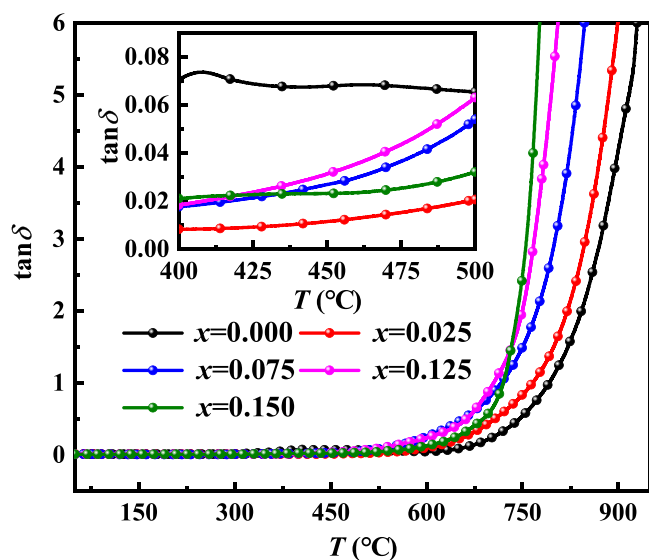


Fig. 9. The relationship between dielectric loss and temperature of CBNCW- x ceramics at 1 MHz.

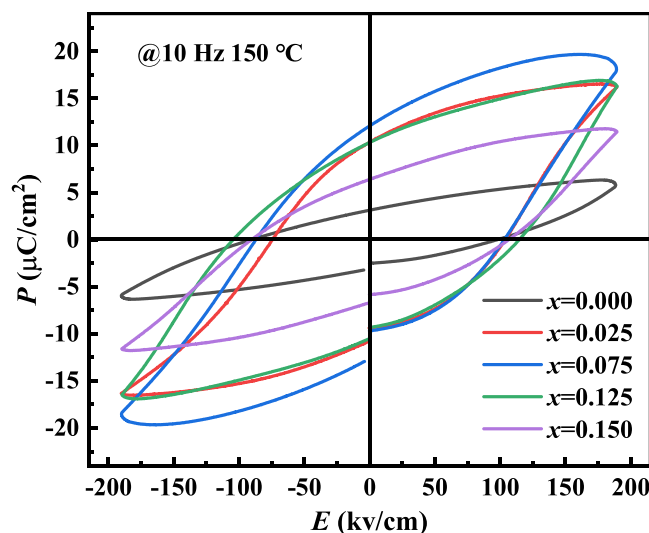


Fig. 10. The hysteresis loops of CBNCW- x ceramic samples.

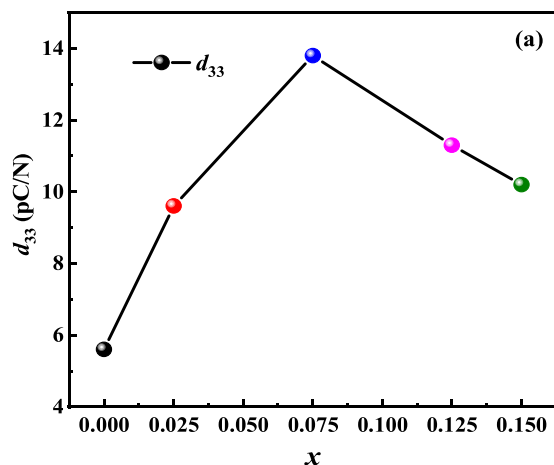


Table 1
The main performances of CBNCW- x ceramic samples at room temperature.

Samples	d_{33} / (pC/N)	P_r / ($\mu\text{C}/\text{cm}^2$)	Q_m	$\tan\delta$ / %
$x = 0.000$	5.6	3.12	550	1.03
$x = 0.025$	9.6	10.3	3702	0.97
$x = 0.075$	13.8	12.1	7099	0.57
$x = 0.125$	11.3	10.1	6044	0.67
$x = 0.150$	10.2	6.39	1 614	0.73

performances of CBNCW- x ceramics, when the doping amount $x = 0.075$, its d_{33} is 13.8 pC/N. The significant improvement of d_{33} is attributed to the improvement of the ferroelectric properties and polarization strength of CBN ceramics by Cu/W ions doping. However, excessive doping causes the reduction of the distortion of the NbO_6 octahedron, the increase of internal defects, and the increased relaxation, which is not conducive to ceramic polarization [39]. Therefore, the appropriate amount of Cu/W ions doping is conducive to the improvement of d_{33} . Fig. 11(b) shows the dependence of the piezoelectric constant d_{33} on the depolarization temperature of the CBNCW- x ceramic samples. In the range of room temperature to 900 °C, the d_{33} of all ceramics declines slightly with temperature, showing good temperature stability. Among them, the d_{33} of the CBNCW-0.075 ceramic at 900 °C is 12.7 pC/N, which only decreases by 7.97% compared with that before depolarization, indicating that it has good high-temperature thermal stability and is suitable for high-temperature environments.

4. Conclusion

The CBNCW- x ceramic samples were prepared by the traditional solid-phase method. All the samples were of a single CBN crystal structure, and the ceramic grains grew well and showed a typical bismuth layered plate-like structure. Cu/W ions doping reduces the degree of orthogonal distortion of CBN ceramics, but an appropriate doping amount can effectively increase the distortion of NbO_6 octahedrons in pseudo-perovskite. In addition, Cu/W doping still shows the effects of additives of CuO, such as effectively reducing the sintering temperature, reducing dielectric loss, and improving quality factor. When the doping amount $x \geq 0.125$, the internal ions are disordered, the relaxation degree increases and the electrical properties decrease rapidly. Therefore, an appropriate amount of doping can greatly improve the piezoelectric and ferroelectric properties while maintaining a high Curie temperature. The high-temperature stability test of all samples performed well. When the doping amount $x = 0.075$, the comprehensive performance of CBN ceramics reaches the best: $d_{33} = 13.8$ pC/N, $P_r = 12.10$ $\mu\text{C}/\text{cm}^2$,

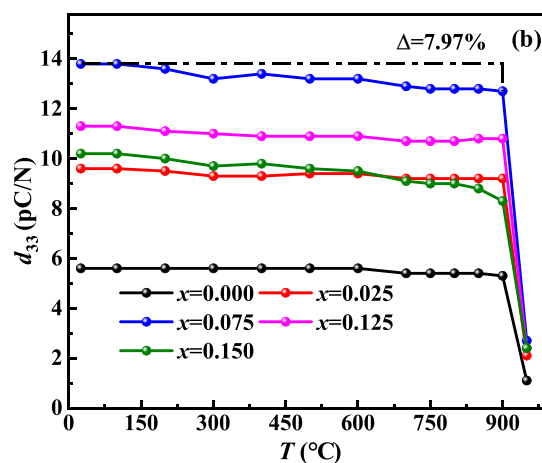


Fig. 11. Piezoelectric properties of CBNCW- x ceramic samples: (a) piezoelectric constant d_{33} and (b) the curve of thermal depoling.

$T_c = 918^\circ\text{C}$, $\tan\delta = 0.57\%$, $Q_m = 7099$. In summary, it is shown that Cu/W ions doping CBN ceramics are an effective modification method and have good high-temperature application potential.

CRedit authorship contribution statement

Hepeng Wang: Conceptualization, Methodology, Visualization, Investigation, Writing – original draft, Writing – review & editing, Formal analysis, Data curation. **Chao Chen:** Conceptualization, Writing – review & editing, Supervision, Formal analysis. **Xiangping Jiang:** Conceptualization, Resources, Writing – review & editing, Supervision, Formal analysis. **Xiaokun Huang:** Conceptualization, Writing – review & editing, Supervision. **Xin Nie:** Formal analysis, Data curation. **Liwei Huang:** Conceptualization, Writing – original draft, Writing – review & editing, Visualization. **Xing Yu:** Supervision, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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